Standard Errors and Confidence Intervals for Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

Ivan Jacob Agaloos Pesigan[1], Michael A. Russell[1, 2], and Sy-Miin Chow[3]

[1]Edna Bennett Pierce Prevention Research Center, The Pennsylvania State University

[2]Department of Biobehavioral Health, The Pennsylvania State University

[3]Department of Human Development and Family Studies, The Pennsylvania State University

Author Note

Ivan Jacob Agaloos Pesigan https://orcid.org/0000-0003-4818-8420; Michael A. Russell https://orcid.org/0000-0002-3956-604X; Sy-Miin Chow https://orcid.org/0000-0003-1938-027X.

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Correspondence concerning this article should be addressed to Ivan Jacob Agaloos Pesigan, Edna Bennett Pierce Prevention Research Center, College of Health and Human Development, The Pennsylvania State University, 320 Biobehavioral Health Building, University Park, PA 16802 or by email (ijapesigan@psu.edu).

Abstract

Mediation modeling using longitudinal data is an exciting field that captures the interrelations in dynamic changes, such as mediated changes, over time. Even though discrete-time vector autoregressive (DT-VAR) approaches are commonly used to estimate indirect effects in longitudinal data, they have known limitations due to the dependency of inferential results on the time intervals between successive occasions and the assumption of regular spacing between measurements. To address these issues, continuous-time vector autoregressive (CT-VAR) models have been proposed as an alternative. Previous work in the area (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) has shown how the direct, indirect, and total effects, for a range of time-interval values, can be calculated using parameters estimated from CT-VAR models for causal inferential purposes. However, methods for calculating the uncertainty around the direct, indirect, and total effects in continuous-time mediation have yet to be explored. Drawing from the mediation model literature, we present and compare results from using the delta and Monte Carlo methods to calculate standard errors and confidence intervals for the direct, indirect, and total effects in continuous-time mediation for inferential purposes. Options to automate these inferential procedures and facilitate interpretations are available in the `cTMed` `R` package.

Keywords: mediation, continuous-time, delta method, Monte Carlo method, R package, cTMed

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Empirical Example

In this section, we demonstrate the application of a continuous-time mediation model using empirical experience sampling data from an individual with a history of major depressive disorder. Specifically, we estimated the CT-VAR drift matrix and the sampling covariance matrix. These estimates allowed us to generate confidence intervals using both the delta method and the Monte Carlo method. Our focus was on examining direct, indirect, and total effects within a model of associations related to depression symptoms.

From a biopsychosocial perspective (inspired by Engel, 1977), we recognize that biology, psychology, and socio-environmental factors are interconnected. Depression, as a mental disorder, manifests with both somatic and emotional symptoms. Somatic symptoms often include physical distress and pain. Research has consistently shown that the relationship between depression and pain is bidirectional: depression predicts the development of chronic pain, and chronic pain increases the risk of developing depression (Von Korff & Simon, 1996). Building on this theoretical framework, we investigated the indirect effect of physical distress on self-esteem. We hypothesized that this indirect pathway would be mediated by psychological distress and negative mood.

Data

In this example, we explore an empirical data set utilized by Ryan and Hamaker (2021), which originates from Wichers et al. (2016) and is further described in Kossakowski et al. (2017). The data set comprises $1,476$ semi-random measurement occasions spanning 239 days. These measurements were collected from a 57-year-old male participant with a history of major depressive disorder. The participant responded to queries multiple times a day. The time intervals between measurement occasions vary, with a minimum of 13.5 minutes and a maximum of 42.1 hours. We focused on the following variables: physical discomfort, psychological distress, negative mood, and self-esteem. For physical discomfort,

we created a composite score for the following items: “I am hungry”, “I am tired”, “I am in pain”, “I feel dizzy”, “I have a dry mouth”, “I feel nauseous”, “I have a headache”, and “I am sleepy”. For psychological distress, we created a composite score for the following items: “I feel restless”, “I feel agitated”, “I worry”, and “I can concentrate well” (reversed). For negative mood, we created a composite score for the following items: “I feel down”, “I feel irritated”, “I feel lonely”, “I feel anxious”, “I feel guilty”, and “I feel indecisive”. Finally, for self-esteem, we created a composite score for the following items: “I like myself”, “I am ashamed of myself” (reversed), “I doubt myself” (reversed), “I can handle anything”. All items were measured using a 7-point Likert scale, and the means of these items were calculated to create the composite scores.

Data Preparation

Before delving into the analysis, we followed a specific data preparation procedure, building upon the approach outlined by (2021). Here are the steps we took: We created a vector of time values by considering the smallest and largest observed values (0 and 5724.554, respectively). These time values were incremented by 0.10. The newly generated time vector was appended to the existing empirical time vector. We sorted the combined time vector and any duplicated time values were removed. If a time value from the new time vector existed in the original data set, we labeled the observations accordingly. For time values not present in the original data set, we inserted missing values.

Data Analysis

We fitted the CT-VAR model using the `dynr` package. We estimated the drift $\boldsymbol{\Phi}$ and diffusion $\boldsymbol{\Sigma}$ matrices. Since the variables were scaled prior to model fitting, we assumed that the centralizing tendency $\boldsymbol{\mu}$ is a a vector of zeros. To simplify the model, we did not estimate the measurement model.

Leveraging the estimated drift matrix $\boldsymbol{\Phi}$ and the corresponding sampling variance-covariance matrix, we calculated direct, indirect, and total effects of the model where the association between physical distress and self-esteem is mediated by

psychological distress and negative mood using the `cTMed` package. We examined the mediated model across a range of time intervals (from > 0 to 15). Confidence intervals were estimated using the delta and Monte Carlo methods.

Results

The estimated CT-VAR parameters are presented in Table 1. The direct and total effects were not significant for most of the time intervals we investigated. The total effect for the Monte Carlo method was significant at the right tail end. The indirect effect as significant for time-intervals greater than 0 to 15 for the Monte Carlo method while it was only significant between a time-interval greater than 1 and less that 8.36 for the delta method. The results are presented in Figure 1.

Discussion

Results showed that the association between physical distress and self-esteem is mediated by psychological distress and negative mood. It is notable that both direct and total effects were not significant for the time-intervals considered suggesting the importance of the mediator variables in the mechanism linking the somatic distress to an individual's overall evaluation of their own worth.

Mislabeled Time-Intervals

We fitted a model where the time-intervals are mislabeled. In this model, instead of considering the actual time observations were made, we assumed that the measurement $1,476$ measurement occasions are equally spaced. As such, we recoded the time variable from 0 to $1,475$. We used the same data preparation and model specification as the correctly specified time-intervals. The CT-VAR drift matrix for this mislabeled model is presented in Table 2. The regions of significance for the direct, indirect, and total effects are presented in Panels g to l of Figure 1. While the pattern of results is similar to the correctly labeled time-intervals, results for the mislabeled model exhibited stronger total effects. In the correctly labeled model, the total effect is not significant across all the time-intervals considered (except for the tail end for the Monte Carlo method), in the

mislabeled case, the total effect starts to be significant at Δt of 0.50 for both delta and Monte Carlo method. The delta method stops to be significant at 8.30 while the Monte Carlo is significant until Δt of 15. In the correctly label model, the indirect effect is significant for delta method between the time-interval greater than 1 and less that 8.36. In the mislabeled model, the delta method is significant from > 0 to 8.30. The Monte Carlo method is significant for all the time-intervals considered for both correctly labeled and mislabeled models.

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Table 1

Point and Interval Estimates of the CT-VAR Drift Matrix $\mathbf{\Phi}$ *for the Empirical Data Set from (Wichers et al., 2016)*

Parameter	Estimate	Standard Error	2.5%	97.5%
Negative mood to negative mood	-0.682	0.091	-0.861	-0.504
Negative mood to psychological distress	-0.256	0.104	-0.460	-0.052
Negative mood to self-esteem	0.041	0.130	-0.215	0.297
Negative mood to physical distress	NA	NA	NA	NA
Psychological distress to negative mood	-0.231	0.101	-0.428	-0.034
Psychological distress to psychological distress	-0.830	0.118	-1.062	-0.598
Psychological distress to self-esteem	-0.030	0.146	-0.317	0.256
Psychological distress to physical distress	NA	NA	NA	NA
Self-esteem to negative mood	-0.155	0.057	-0.267	-0.043
Self-esteem to psychological distress	0.005	0.066	-0.124	0.135
Self-esteem to self-esteem	-0.898	0.081	-1.057	-0.740
Self-esteem to physical distress	NA	NA	NA	NA
Physical distress to negative mood	NA	NA	NA	NA
Physical distress to psychological distress	NA	NA	NA	NA
Physical distress to self-esteem	NA	NA	NA	NA
Physical distress to physical distress	NA	NA	NA	NA

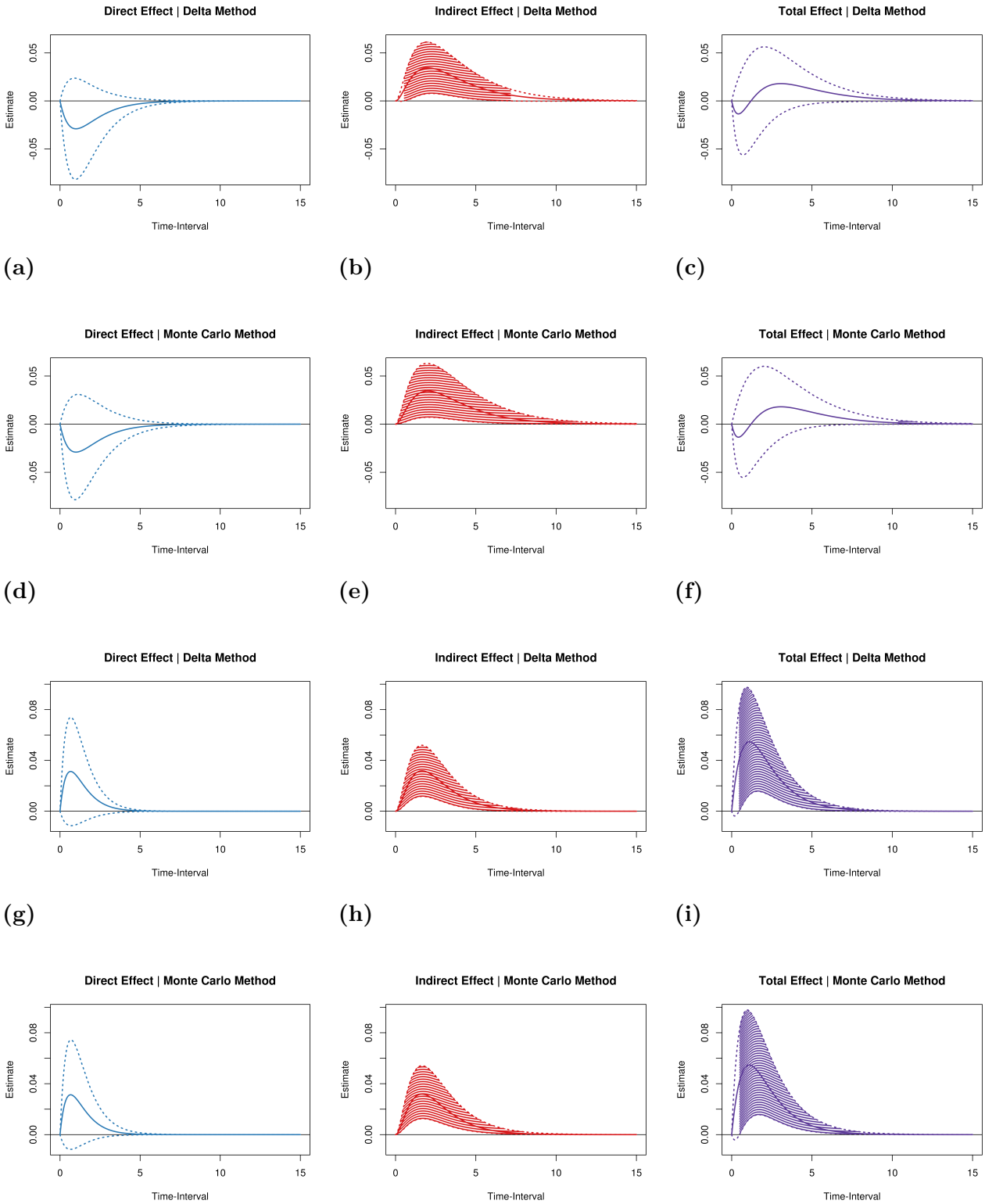
Table 2

Point and Interval Estimates of the CT-VAR Drift Matrix $\mathbf{\Phi}$ *for the Empirical Data Set from (Wichers et al., 2016) with Mislabeled Time-Intervals*

Parameter	Estimate	Standard Error	2.5%	97.5%
Negative mood to negative mood	-1.067	0.121	-1.305	-0.829
Negative mood to psychological distress	-0.339	0.128	-0.590	-0.088
Negative mood to self-esteem	0.041	0.182	-0.315	0.397
Negative mood to physical distress	NA	NA	NA	NA
Psychological distress to negative mood	-0.395	0.130	-0.651	-0.140
Psychological distress to psychological distress	-1.136	0.137	-1.405	-0.866
Psychological distress to self-esteem	-0.014	0.196	-0.397	0.370
Psychological distress to physical distress	NA	NA	NA	NA
Self-esteem to negative mood	-0.326	0.085	-0.492	-0.161
Self-esteem to psychological distress	0.195	0.090	0.019	0.370
Self-esteem to self-esteem	-1.570	0.128	-1.820	-1.319
Self-esteem to physical distress	NA	NA	NA	NA
Physical distress to negative mood	NA	NA	NA	NA
Physical distress to psychological distress	NA	NA	NA	NA
Physical distress to self-esteem	NA	NA	NA	NA
Physical distress to physical distress	NA	NA	NA	NA

Figure 1

Regions of Significance for the Direct, Indirect, and Total Effects (Physical Distress → Psychological Distress and Negative Mood → Self-Esteem)



Note: Panels 1a to 1f correspond to the model where the time-intervals are correctly labeled. Panels 1g to 1l correspond to the model where the time-intervals are incorrectly label.